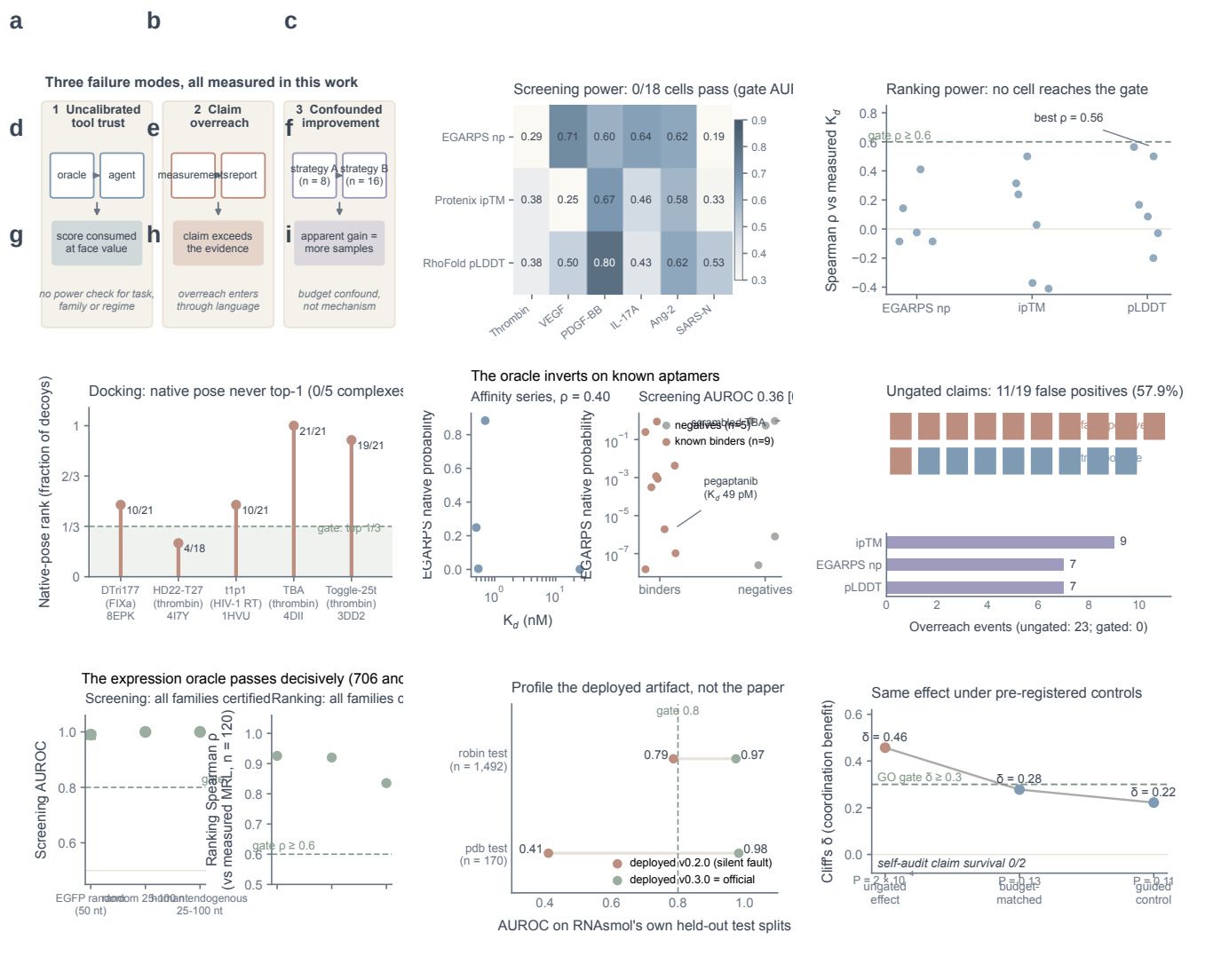

a
b
c
Three failure modes, all measured in this work
d
1 Uncalibrated tool trust
oracle
agent
g
score consumed at face value
no power check for task, family or regime
e
2 Claim overreach
measurement
report
h
claim exceeds the evidence
overreach enters through language
f
3 Confounded improvement
strategy A (n = 8)
strategy B (n = 16)
i
apparent gain = more samples
budget confound, not mechanism
Screening power: 0/18 cells pass (gate AUI
EGARPS np
Protenix ipTM
RhoFold pLDDT
0.29
0.71
0.60
0.64
0.62
0.19
0.38
0.25
0.67
0.46
0.58
0.33
0.38
0.50
0.80
0.43
0.62
0.53
Thrombin
VEGF
PDGF-BB
IL-17A
Ang-2
SARS-N
Ranking power: no cell reaches the gate
Spearman ρ vs measured K_d
best ρ = 0.56
gate ρ ≥ 0.6
EGARPS np
ipTM
pLDDT
Docking: native pose never top-1 (0/5 complexes
Native-pose rank (fraction of decoys)
10/21
4/18
10/21
21/21
19/21
gate: top 1/3
DTri177 (FIXa) 8EPK
HD22-T27 (thrombin) 4I7Y
t1p1 (HIV-1 RT) 1HVU
TBA (thrombin) 4DII
Toggle-25t (thrombin) 3DD2
The oracle inverts on known aptamers
Affinity series, ρ = 0.40
Screening AUROC 0.36 [
EGARPS native probability
K_d (nM)
negatives (n=5)
known binders (n=9)
pegaptanib (K_d 49 pM)
binders
negatives
Ungated claims: 11/19 false positives (57.9%)
ipTM
EGARPS np
pLDDT
9
7
7
Overreach events (ungated: 23; gated: 0)
The expression oracle passes decisively (706 anc
Screening: all families certified
Ranking: all families c
Screening AUROC
Ranking Spearman ρ (vs measured MRL, n = 120)
gate
gate ρ ≥ 0.6
EGFP random (50 nt)
random 25-100 nt
endogenous 25-100 nt
Profile the deployed artifact, not the paper
gate 0.8
robin test (n = 1,492)
pdb test (n = 170)
0.79
0.97
0.41
0.98
deployed v0.2.0 (silent fault)
deployed v0.3.0 = official
AUROC on RNAsmol's own held-out test splits
Same effect under pre-registered controls
Cliff's δ (coordination benefit)
δ = 0.46
δ = 0.28
δ = 0.22
GO gate δ ≥ 0.3
self-audit claim survival 0/2
P = 2 × 10⁻⁴
P = 0.13
P = 0.11
ungated effect
budget-matched
guided control